

Implementation Plan

Evaluating Explanation Consistency in Zero-Shot LLM Toxicity Classification

The implementation will be divided across three participants. Each participant owns one major part of the pipeline, while the final integration will combine classification, explanation, comparison, and reporting outputs.

Participants should agree on and lock the output file schemas at the end of Day 1 so that downstream work can proceed independently. In particular, Participant 2 depends on Participant 1’s output file, and Participant 3 depends on outputs from both. Participant 3 should define and mock expected input schemas on Day 1 so that comparison and fairness logic can be developed in parallel.

1 Participant 1: Dataset Preparation and Zero-Shot Classification

Participant 1 will implement the data loading and baseline classification pipeline.

- Load the Civil Comments dataset and select the required fields: `comment_text`, `target`, and available identity columns.
- Normalize the dataset and create a capped working subset for experimentation.
- Convert the continuous toxicity target into a binary label using a selected threshold, such as `target > 0.5`.
- Implement the zero-shot toxicity classification prompt.
- Compute the model score:

$$s(x) = \log p(\text{toxic} \mid x) - \log p(\text{non-toxic} \mid x)$$

- Generate predictions using:

$$\hat{y} = \begin{cases} 1 & \text{if } s(x) > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Save the main prediction output file containing: `idx`, `comment_text`, `target`, `binary_label`, `log_prob_toxic`, `log_prob_nontoxic`, `score`, and `prediction`.

Note: both log-probability components must be saved separately, not only the combined score $s(x)$, so that downstream analyses can reconstruct confidence and uncertainty estimates.

2 Participant 2: Explanation Signal Extraction

Participant 2 will implement the two explanation methods used for comparison: Integrated Gradients and attention-based token importance.

- Select a smaller comparison subset from the prediction output, since explanation methods are more computationally expensive. Record the selected `idx` values so Participant 3 can align results.
- Implement Integrated Gradients on the toxicity score $s(x)$, using:

- zero embedding as the baseline,
- linear interpolation from the baseline to actual input embeddings,
- a fixed number of interpolation steps (e.g., 32),
- gradient computation through `inputs_embeds` to ensure attribution reflects prompt-token contributions.

Positive attribution indicates contribution toward *toxic*; negative attribution indicates contribution toward *non-toxic*.

- For each selected example, compute token-level IG attribution scores.
- Extract attention weights from selected transformer layer(s), such as the final layer or final few layers.
- Aggregate attention weights across heads into a single token-level attention importance vector.
- Normalize IG and attention scores so that they can be compared over the same token sequence.
- Save an explanation output file containing: `idx`, `tokens`, `ig_scores`, and `attention_scores`.
- On Day 3, produce three target heatmaps showing IG and attention side by side:
 1. a correctly classified toxic example with high IG–attention agreement,
 2. a false positive where the two methods diverge,
 3. a borderline or ambiguous example illustrating how disagreement relates to model uncertainty.

3 Participant 3: Comparison, Fairness Metrics, and Reporting

Participant 3 will implement the comparison logic, fairness evaluation, correlation analysis, and final reporting outputs.

- Align the IG attribution vector and attention importance vector for each selected example.
- Compute explanation agreement metrics, such as top- k token overlap or rank correlation, producing a per-example agreement score.
- Identify examples where IG and attention agree strongly, and examples where they diverge.
- Analyze whether explanation disagreements are correlated with borderline predictions, fairness-relevant identity terms, or elevated model uncertainty. This correlation analysis requires merging the explanation agreement scores with the prediction scores and identity columns from Participant 1’s output.
- Merge classification predictions with identity columns.
- Apply a minimum group-size filter (e.g., $n \geq 50$) before computing per-group disparity metrics, to avoid unstable small-sample estimates.
- Compute per-group fairness metrics for each identity subgroup ($A = 0$ and $A = 1$), including: accuracy, F1, true positive rate (TPR), false positive rate (FPR), and positive prediction rate.
- Compute disparity metrics for each identity:

$$\text{SPD} = P(\hat{Y} = 1 \mid A = 1) - P(\hat{Y} = 1 \mid A = 0) \quad (1)$$

$$\text{EOpp} = \text{TPR}_{A=1} - \text{TPR}_{A=0} \quad (2)$$

- Report worst-case metrics across all identities: $\max |\text{SPD}|$, $\max |\text{EOpp}|$, worst subgroup accuracy, and worst subgroup F1.
- Produce final output files:

- `predictions.csv` — prediction results,
- `explanation_comparison.csv` — per-example IG and attention scores,
- `explanation_agreement.csv` — per-example agreement score and divergence flag (`idx`, `agreement_score`, `divergence_flag`),
- `fairness_metrics.csv` — per-group and per-identity metrics,
- `worst_case_summary.csv` — worst-case subgroup summary,
- selected heatmap visualizations (produced by Participant 2).

4 Three-Day Build Schedule

4.1 Day 1: Pipeline Skeleton and Schema Lock

The goal of Day 1 is to get the basic end-to-end structure working and to lock inter-participant file schemas so downstream work can proceed independently.

- Participant 1 loads the dataset and produces the first prediction file, including all required columns (`idx`, `comment_text`, `target`, `binary_label`, `log_prob_toxic`, `log_prob_nontoxic`, `score`, `prediction`).
- Participant 2 creates placeholder functions for IG and attention extraction, and confirms the explanation output schema (`idx`, `tokens`, `ig_scores`, `attention_scores`).
- Participant 3 defines the expected output format for comparison and reporting, and mocks the input schemas so that fairness and agreement logic can be developed in parallel.

4.2 Day 2: Explanation Methods

The goal of Day 2 is to replace placeholder logic with real explanation signals.

- Participant 1 improves or validates the zero-shot scoring logic.
- Participant 2 implements real IG and attention extraction on a small subset.
- Participant 3 implements top- k overlap or rank correlation between explanation methods, and begins the fairness metric pipeline.

4.3 Day 3: Analysis and Presentation

The goal of Day 3 is to produce interpretable results and prepare the final presentation.

- Participant 1 finalizes classification results.
- Participant 2 finalizes explanation outputs and produces three target heatmaps: (1) a correctly classified toxic example with high IG–attention agreement, (2) a false positive where the two methods diverge, and (3) a borderline example illustrating how disagreement relates to model uncertainty.
- Participant 3 finalizes fairness metrics, agreement metrics, the divergence correlation analysis, and all summary findings and output CSVs.

5 Minimum Viable Deliverable

The minimum viable deliverable is an end-to-end prototype that:

1. loads a sample of Civil Comments,
2. classifies comments as toxic or non-toxic using zero-shot scoring,

3. computes IG and attention scores for a small subset,
4. compares the two explanation signals and produces per-example agreement scores,
5. produces at least one side-by-side heatmap,
6. and reports basic fairness metrics across identity groups.